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Microglia Sculpt Sex Differences in Social Behavior

Caroline J. Smith^{1,2} and Staci D. Bilbo^{1,2,*}

¹Lurie Center for Autism, Massachusetts General Hospital for Children, Boston, MA 02114, USA

²Department of Pediatrics, Harvard Medical School, Boston, MA 02114, USA

*Correspondence: sbilbo@mgh.harvard.edu

<https://doi.org/10.1016/j.neuron.2019.03.039>

Microglia are increasingly recognized as developmental sculptors of neural circuits. In this issue of *Neuron*, VanRyzin et al. (2019) demonstrate a novel mechanism by which endocannabinoids drive microglia to phagocytose newborn astrocytes in the medial amygdala of male rats, promoting sex differences in social play behavior.

Sex is a biological variable that influences the expression of many social behaviors across the animal kingdom. These behaviors include adult reproductive behaviors, such as mating, parental behavior, and aggression, but also behaviors that emerge earlier in development, such as adolescent social play behavior. The neonatal testosterone (T) surge, which occurs in male rodents around the day of birth, organizes many of these sex differences in social behavior. However, much less is known regarding the mechanisms by which changes in T lead to changes in the neural circuitry that supports social behavior. Microglia, the resident macrophages of the brain, exhibit sex differences in their morphology, suggesting sex differences in their behavior, during the perinatal period (Schwarz et al., 2012). Moreover, they play a critical role in the refinement of neural circuits, e.g., within the developing visual system (Stevens et al., 2007; Schafer et al., 2012). Only very recently have microglia been demonstrated to play a role in the developmental organiza-

tion of social circuits and, thereby, to impact the behaviors those circuits subserve (Kopec et al., 2018). In this issue of *Neuron*, VanRyzin et al. (2019) provide the first evidence for a mechanistic pathway by which the neonatal T surge drives microglia to differentially sculpt social circuits in males as compared to females.

VanRyzin et al. (2019) began with the interesting observation that male rats have higher numbers of phagocytic microglia (defined as having one or more phagocytic “cups” at the distal ends of processes) than females in the amygdala between birth and postnatal day 4 (P4), but not later in development. They observed no sex differences in the total number of microglia or in other aspects of microglial morphology, i.e., ramification. To query the contents of the phagocytic cups and address the question of what microglia are eating, VanRyzin et al. (2019) used Imaris-reconstructed immunohistochemistry. In both males and females, the majority of phagocytic cups contained the nuclei of newly proliferating cells, sug-

gesting that these microglia were engulfing newborn cells. Concordantly, only a small minority of cups contained a marker for apoptotic cells.

Together, these findings suggest that male microglia phagocytose higher numbers of newborn cells in the developing amygdala than female microglia. Indeed, this is in line with previous work from the McCarthy lab demonstrating that males have fewer newborn cells (BrdU-labeled) in the amygdala at P4 than females. Moreover, treatment of female pups with endocannabinoid receptor agonists at P0 and P1 masculinized both newborn cell number and the expression of social play behavior (Krebs-Kraft et al., 2010). Based on the above findings, VanRyzin et al. (2019) hypothesized that neonatal T programs microglia to induce sex differences in social play behavior by phagocytosing newborn cells. They also hypothesized that this process is mediated, at least in part, by endocannabinoid signaling.

Next, VanRyzin et al. (2019) investigated whether neonatal T treatment in females

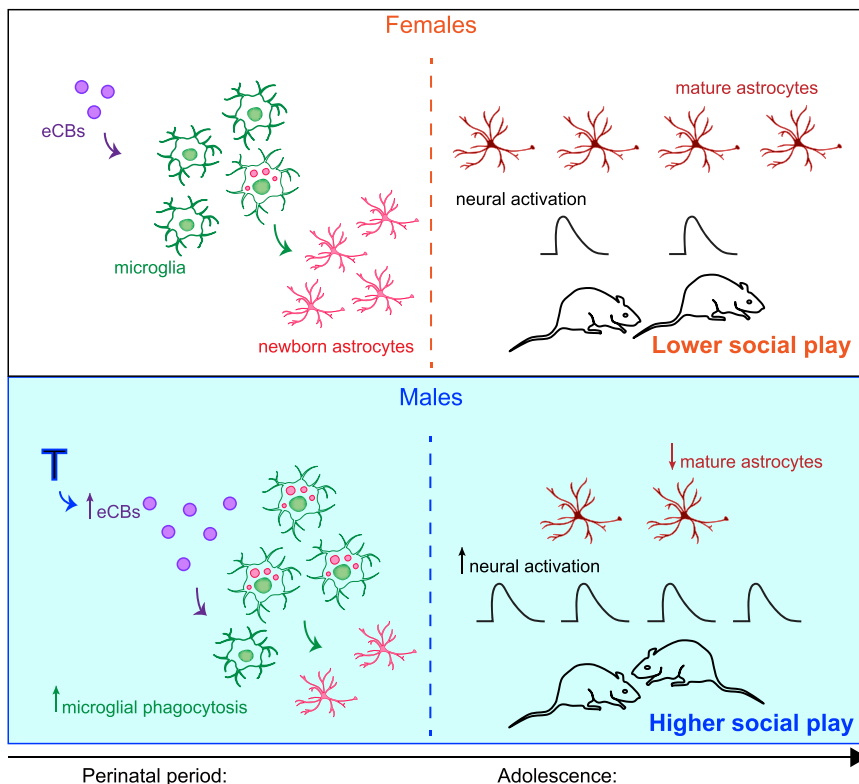


Figure 1. Microglial Phagocytosis of Newborn Astrocytes within the Developing Amygdala Leads to Sex Differences in Social Play Behavior

In males as compared to females, the neonatal testosterone surge increases endocannabinoid tone within the amygdala, which then increases microglial phagocytosis of newborn astrocytes. As a result, mature astrocyte number is higher in males than in females during adolescence, while neural activation is lower. Causal manipulation studies reveal that both endocannabinoid signaling and microglial phagocytosis are required for the higher levels of social play behavior observed in males. T, testosterone; eCB, endocannabinoid.

could masculinize microglial phagocytosis. They found that T-treated females were indistinguishable from males in both number of phagocytic microglia and number of newborn cells within the amygdala. Intriguingly, they also found that females had lower levels of the endocannabinoid ligand 2-arachidonylglycerol (2-AG) than both males and T-treated females. Furthermore, treatment with agonists against endocannabinoid receptors mimicked the effects of T on both microglial phagocytosis and newborn cell number in females, possibly suggesting that T increases microglial phagocytosis by increasing endocannabinoid tone within the amygdala. Finally, T-treated females exhibited male-like levels of social play behavior during adolescence but not if they were concurrently treated with endocannabinoid receptor antagonists. It remains unclear whether endocannabinoid receptor antag-

onism would prevent the masculinizing effects of T on microglial phagocytosis. However, these findings strongly support a working model in which T serves to increase both microglial phagocytosis of newborn cells and social play behavior in males by increasing endocannabinoid tone in the amygdala. Such a role for endocannabinoid signaling in neurodevelopmental processes is especially important to understand given that it might suggest maternal cannabis use to be particularly risky during the perinatal period.

The previous data clearly demonstrate sex differences in both social play (higher in males) and microglial phagocytosis of newborn cells (also higher in males). But is there a causal relationship between increased microglial phagocytosis of newborn cells and higher social play behavior in males? Microglia identify and engulf some of their phagocytic targets in

a complement-dependent manner (Stevens et al., 2007; Schafer et al., 2012; Kopec et al., 2018). To address the question of causality, VanRyzin et al. (2019) administered a function-blocking antibody against the CD11b component of complement receptor 3 (C3) to inhibit microglial phagocytosis. Anti-CD11b administration into the amygdala at P0 and P2 increased the number of newborn cells at P4 in males only. Importantly, this effect also seems to be mediated by endocannabinoid signaling because it could be prevented by endocannabinoid receptor antagonism in males and induced by endocannabinoid receptor agonism in females. This blockade of microglial phagocytosis also decreased the levels of male social play behavior to that of females.

The finding that microglia organize social circuits by complement-mediated phagocytosis is in line with the findings of another recent study. Kopec et al. (2018) found that microglial, complement-dependent phagocytosis of dopamine D1 receptors in the nucleus accumbens organizes the developmental trajectory of social play behavior in male rats only. Together, these studies provide compelling evidence for a role of the complement system in the neural sculpting of social circuits. This may be highly relevant for disorders such as autism that are characterized by both microglial abnormalities and deficits in social functioning.

They also open the door to new avenues of research into the specific ways in which the complement system instructs microglial pruning. In the current study, flow cytometry revealed that a higher proportion of newly proliferating cells are also positive for the complement proteins C1q and C3 as compared to older cells. However, in mice lacking CD47, microglia excessively and indiscriminately prune synapses during neural development, suggesting that there may also be important “don’t eat me” signals in this brain region (Lehrman et al., 2018).

Given that these mechanisms appear to be driven by testosterone and are male biased, in future studies it will be important to elucidate the role of microglia in the organization of *female-biased* social behaviors and/or disorders and their underlying mechanisms.

Finally, VanRyzin et al. (2019) asked how a decrease in the number of newborn

amygdala cells that survive to adulthood might lead to a change in social play behavior. Their results suggest that male microglia preferentially phagocytose newborn astrocytes and that the resulting decrease in astrocytes in males is accompanied by an increase in markers of neuronal activation. Hence, they postulate that male microglia phagocytose more newborn astrocytes, thereby inducing greater neuronal activation in the medial amygdala and increasing social play behavior (Figure 1). It remains to be determined whether the increase in neural activation within the medial amygdala is causal to the increase in social play. This seems likely, however, given that the medial amygdala is known to regulate social play and sends projections to other brain regions that support social play, such as the lateral septum. Indeed, a previous study also observed higher immediate early gene activation in the medial amygdala during social play, albeit more so in females than in males (Reppucci et al., 2018).

Many interesting questions remain open and are fertile ground for future experiments. For example, what cell types modulate endogenous 2-AG tone within the neonatal amygdala? While microglia are capable of producing endocannabinoids, microglial depletion failed to reduce 2-AG content in the amygdala (VanRyzin et al., 2019). Astrocytes can also regulate endocannabinoid tone and addition of astrocytes to cell culture upregulates the expression of C1q in retinal ganglion cells (Stevens et al., 2007). This may suggest the intriguing possibility that astrocytes regulate the survival of their own kind.

Another question is regarding the specificity of the brain region/circuit mechanisms identified. In the hippocampus, female microglia appear to be more phagocytic than male microglia (Nelson et al., 2017). Microglia preferentially clear apoptotic cells within the cerebellum and epigenetic programming prevents this clearance from occurring within the striatum (Ayata et al., 2018). In the present study, VanRyzin et al. (2019) assessed other nodes of the social reward circuit and found no evidence of sex differences in microglial phagocytosis. Thus, it would be of interest to determine what neural mechanisms might confer such brain region specificity.

In sum, the findings herein elegantly delineate a novel mechanism underlying sex differences in social play behavior and set the stage for future studies investigating the role of microglia in the organization of sexually dimorphic neural circuits, particularly those that mediate social behavior. This is especially critical given that many neuropsychiatric disorders are characterized by changes in social behavior, a sex bias in prevalence, and microglial dysfunction.

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